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Subject : DBHS

Assignment-2

Section-A

Q1 Fill in the blanks

1. The _____ visually represents how entities relate in a database, offering a clear blueprint of the data's organisational structure
→ Entity Relationship (ER) Model

2. In the _____ model, data is organized into tables, with relationships between entities defined through foreign Key Constraint. → Relational

3. The _____ model treat data as a collection of objects, each with unique attributes and methods, mirroring real-world entities and interactions. → Object-oriented

4. _____ constraints ensure the reliability of data by enforcing rules that must be adhered to, maintaining the accuracy of information
→ Integrity

Section-B (Case Study based)

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Q2 In a healthcare system, discuss how data abstraction can simplify the interaction between healthcare professionals and patient records.

→ In a healthcare System, data abstraction means hiding complex databases details and showing only the information that a user needs. This makes it easier for healthcare professionals to interact with patient records.

The data abstraction helps in a way that:-

- It simplifies information Access
- And Display only Role-Based information
- It Reduces Complexity
- It Improves data Security
- And Easier System Maintenance.

Q3 Explore the potential benefits and challenges associated with implementing data abstraction in this context.

→ Benefits

1. Ease of Use: Healthcare professionals can easily access patient information without understanding complex database structures.
2. Improved Security : Sensitive patient data is protected because users only see the information they are authorized to access.

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3 Reduced Complexity: The System hides complicated database operations and shows only necessary data.

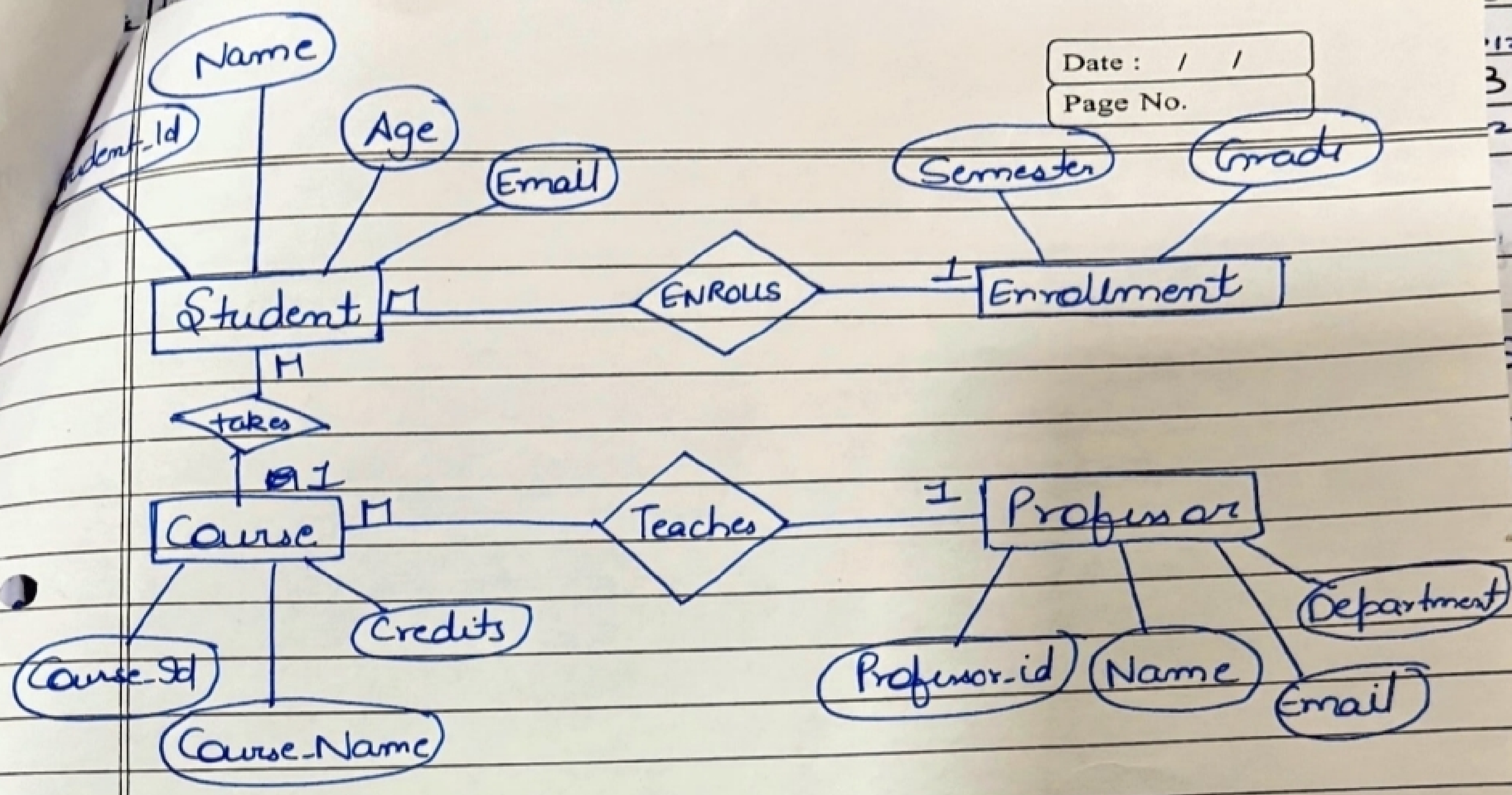
challenges

1. System Design Complexity - Creating proper abstraction layers can be difficult and requires careful system design.
2. The challenge which can occur be performance issues.
3. updating the system or fixing the errors can be a challenge for maintenance.

Section- C (Simulation Based)

Q4 Design a database for a University, incorporating entities like Students, Courses and professors. Using the E-R model, illustrate and explain the relationships between these ~~two~~ entities. Include attributes for each entity to capture relevant information such as Student names, Course details and professor information.

→



Here the relationship is between Student and Enrollement whose Cardinality is Many to One because Many Student can enrolls and a relation between Student and Course that is Many to One because Many Student can takes one course and a relationship between Course and professor which is Many to one because a professor can teach many Courses.